

Question:

$\mathbf{P}$ and $\mathbf{R}$ are Jacobian matrices for two different geophysical inverse problems. If their generalized inverses are written as $\mathbf{P}^{-g} = (\mathbf{P}^T \mathbf{P})^{-1} \mathbf{P}^T$ and $\mathbf{R}^{-g} = \mathbf{R}^T (\mathbf{R} \mathbf{R}^T)^{-1}$, then

- 1) $\mathbf{P}$ represents an overdetermined problem and $\mathbf{R}$ represents an underdetermined problem
- 2) $\mathbf{P}$ represents an underdetermined problem and $\mathbf{R}$ represents an overdetermined problem
- 3) Both $\mathbf{P}$ and $\mathbf{R}$ represent overdetermined problems
- 4) Both $\mathbf{P}$ and $\mathbf{R}$ represent underdetermined problems

Solution:

Given are two Jacobian matrices $\mathbf{P}$ and $\mathbf{R}$ with generalized inverses:

$$\mathbf{P}^{-g} = (\mathbf{P}^T \mathbf{P})^{-1} \mathbf{P}^T \quad (4.1)$$

$$\mathbf{R}^{-g} = \mathbf{R}^T (\mathbf{R} \mathbf{R}^T)^{-1} \quad (4.2)$$

Let $\mathbf{P} \in \mathbb{R}^{m \times n}$ and $\mathbf{R} \in \mathbb{R}^{m \times n}$

$$\mathbf{P}^{-g} = (\mathbf{P}^T \mathbf{P})^{-1} \mathbf{P}^T \quad (4.3)$$

$$\mathbf{P}^T \mathbf{P} \in \mathbb{R}^{n \times n} \quad (4.4)$$

For the inverse to exist

$$\det(\mathbf{P}^T \mathbf{P}) \neq 0 \quad (4.5)$$

$$\implies \text{rank}(\mathbf{P}) = n \leq m \quad (4.6)$$

This implies $m \geq n$, i.e., more equations than unknowns. Hence, $\mathbf{P}$ is tall and the system is overdetermined

$$\mathbf{R}^{-g} = \mathbf{R}^T (\mathbf{R} \mathbf{R}^T)^{-1} \quad (4.7)$$

$$\mathbf{R} \mathbf{R}^T \in \mathbb{R}^{m \times m} \quad (4.8)$$

For the inverse to exist

$$\det(\mathbf{R} \mathbf{R}^T) \neq 0 \quad (4.9)$$

$$\implies \text{rank}(\mathbf{R}) = m \leq n \quad (4.10)$$

This implies $n \geq m$, i.e., more unknowns than equations. Hence, $\mathbf{R}$ is wide and the system is underdetermined

$\mathbf{P}$ represents an overdetermined system and $\mathbf{R}$ represents an underdetermined system